

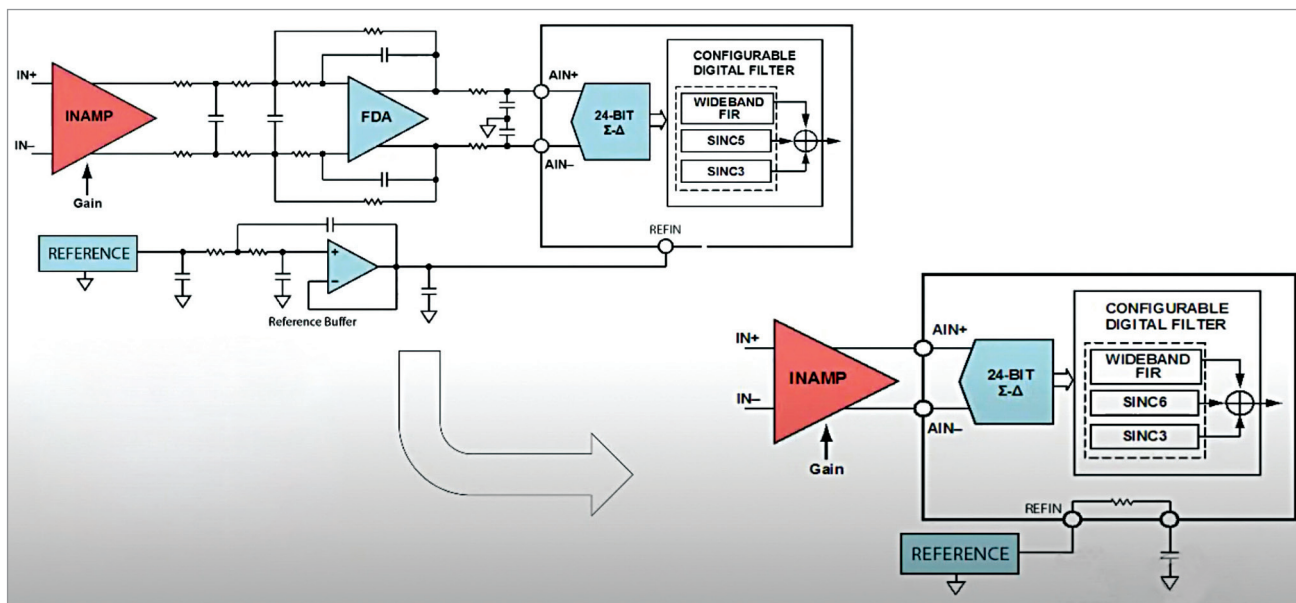


Fig. 5: Simplified ADC design

TABLE 1
MINIMUM AND MAXIMUM SPECIFICATIONS FOR ADCs

ADC Architecture/Specs	Sample Rate/Speed	Resolution/Bits	Price	Power
SAR	Up to 15Msps	8 to 24	Low to medium	Lowest per kspcs
Sigma Delta	Up to 20Msps*	12 to 32	Low to medium	Low to medium
Pipeline	10Msps to 5Gsps	8 to 16	Highest	Highest
μModules	DC to 1Msps	16 to 24	Medium to high	Low to medium

*Modulator output speed

the modulator frequency rate is high, the noise inline gets spread across large bandwidth. Further, noise shaping is used to shift noise out of the band of interest, which can then be removed with low pass digital filtering. Such ADCs are very popular when high resolution is desired and find application in slow-moving input signals like temperature and pressure.

Staying on top of the tech

A lot of interesting innovations are happening in this space, which could in turn do a lot of good to your application too. Integration and simplification seem to be the most effective mantras today.

Designers are trying to integrate single-chain components like amplifiers and filters on the ADC

chip itself. All discrete implementations have been moved onto a single IC, called a micro-module. This can help reduce board space significantly.

The continuous time sigma delta ADC is another new development worth watching, as this architectural approach offers major benefits of inherent anti-aliasing and resistive inputs.

In traditional signal chains there would be a requirement of high bandwidth driver to drive the switch capacitor input and low pass filter to make sure Nyquist is met and there is no aliasing. Also, there would be a reference buffer needed to drive the switched cap reference input. This adds to components and increases the bill of materials.

The continuous time sigma delta ADC has a resistive input and the sampling part is introduced at a much later stage, keeping the entire loop continuous. This removes the requirement of the high bandwidth driver, which is already present to solve the sample problem with it. The other benefit of this continuous loop is that any aliasing that gets introduced automatically gets rejected. Hence, the requirement of high-bandwidth driver and low-pass filter is totally removed from the signal chain, which reduces number of components, simplifies design, and makes bill of materials management simple.

It is recommended for designers to use manufacturers' selection guides along with some basics on architecture types. The designers can use tools, devices, and design aids to support the selection process. Do not overlook input, output signals and consult data sheets or application notes, if any. As ADC technologies improve, it will lead the way for efficient electronics devices. **EFY**

The article is based on a talk by Wasim Shaikh, Product Applications Manager, Analog Devices, at the Design Unconference.